University of Heidelberg

Faculty of Mathematics and Computer Science
Institute of Computer Science
Research Group: Data Analysis and Modeling in Medicine

Bachelor's Thesis in Computer Science

Data-driven camera calibration for light field camera systems

Name: Youssef Daoudi El Boukhrissi
Student ID: 4277022
Supervisor: Prof. Dr. Jürgen Hesser
Submission date: 28 February 2026

Abstract

Contents

List of Figures

1. Introduction

1.1. Motivation

1.2. Challenges and Problem Statement

1.3. Research Questions

The following research questions guide this thesis:

- **RQ1:** Can the target-free calibration method LiFCal be used to automatically calibrate a multi-camera light field system without human interaction or the use of a checkerboard?
- **RQ2:** How well does LiFCal perform when calibrating a multi-camera light field system, and can it replace a classical checkerboard-based calibration?
- **RQ3:** Under which conditions and requirements can an automatic, target-free recalibration of a multi-camera light field system be successfully performed?

1.4. Structure of the Thesis

2. Theoretical Background

2.1. Pinhole Camera Model

Cameras are the primary devices used to capture visual information from the real world into an image. To process and analyze this information computationally, it is necessary to describe the imaging process using a camera model [3].

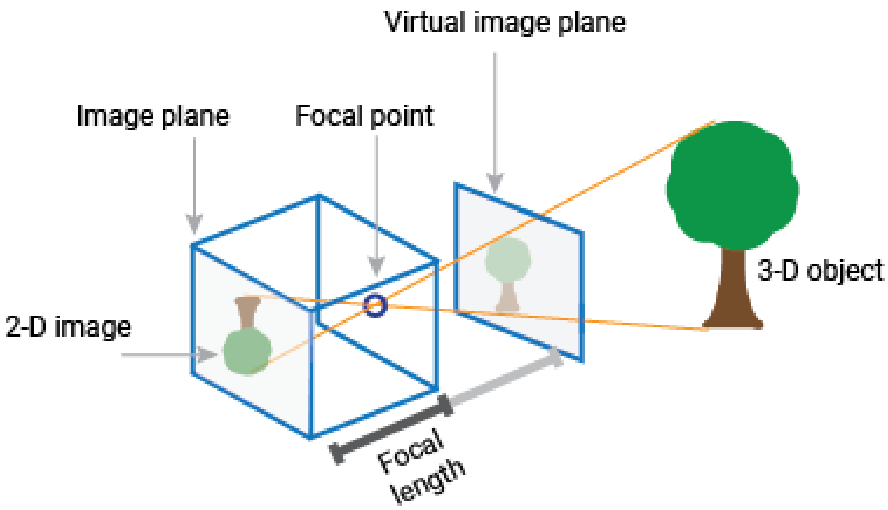

Figure 1: Illustration of the pinhole camera model. Source: [5].

The pinhole camera model describes a simple imaging system in which light rays from a 3D scene pass through a small aperture and are projected through the focal point onto a 2D image plane [3]. The aperture limits the incoming light such that each point in the scene is mapped to a single point on the image plane. In this model, the projected image is inverted due to the geometry of the projection, as shown in Figure 1.

To define the mapping from 3D world coordinates to 2D image coordinates, the model is mathematically described using two types of parameters [5].

- **Intrinsic Parameters:**

$$K = \begin{bmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}$$

  where (f_x, f_y) are the focal lengths in pixel units along the x and y axes, s is the skew coefficient between the axes, and (c_x, c_y) is the principal point (optical center) in pixel coordinates. K is also known as the camera matrix.

- **Extrinsic Parameters:**

$$[R \mid t] = \begin{bmatrix} r_{11} & r_{12} & r_{13} & t_x \\ r_{21} & r_{22} & r_{23} & t_y \\ r_{31} & r_{32} & r_{33} & t_z \end{bmatrix}$$

where R is a 3×3 rotation matrix representing the orientation of the camera with respect to the world coordinate system, and t is a 3×1 translation vector representing the position of the camera center in world coordinates.

These parameters form the basis of camera calibration, which aims to estimate them accurately. However in practice, cameras use lenses to focus light onto the image sensor, which introduces distortions not accounted for in the idealized pinhole model.

2.2. Lens Distortion

The camera matrix alone is not sufficient to model real imaging systems, as the ideal pinhole camera model assumes a lens-free projection. In practice, however, camera lenses introduce various optical distortions that must be taken into account to accurately describe the image formation process [5].

Lens distortion is defined as an optical aberration that causes straight lines in the real world to appear curved in the captured image [2]. Since distortion alters the geometric relationship between 3D world points and their 2D image projections, it has a direct impact on the accuracy of camera calibration. If lens distortion is not properly modeled, the estimated intrinsic and extrinsic parameters can become biased, leading to unreliable calibration results [7].

In most camera calibration models, lens distortion is represented using a small set of parametric functions. The two most common types of distortion considered in practice are radial distortion and tangential distortion.

2.2.1. Radial Distortion

Radial distortion is caused by the spherical shape of camera lenses, which leads to different refraction of light from the image center to the outer regions. This causes straight lines in the image to appear curved, resulting in barrel or pincushion distortion [1].

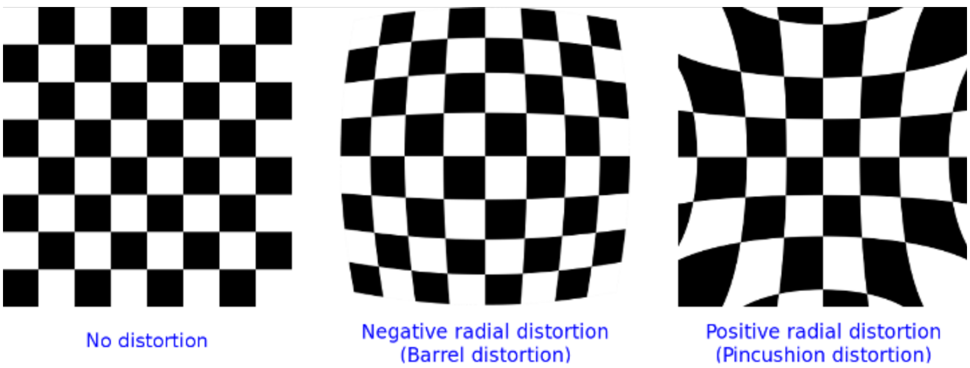

Figure 2: Comparison of different types of radial distortion. Source: [6].

The formal definition of radial distortion is given as follows [5]:

$$x_{distorted} = x\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) \quad (1)$$

$$y_{distorted} = y\left(1 + k_1 r^2 + k_2 r^4 + k_3 r^6\right) \tag{2}$$

where x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \tag{3}$$

The coefficients k_1, k_2, and k_3 in (1) and (2) control the strength of radial lens distortion. In practice, the first coefficient k_1 has the dominant influence on the distortion type: negative values of k_1 typically result in barrel distortion, while positive values lead to pincushion distortion.

2.2.2. Tangential Distortion

Tangential distortion occurs when the image sensor and the camera lens are not perfectly aligned or parallel to each other [5]. As a result, image points are displaced tangentially with respect to the optical axis. Figure 3 illustrates the geometric cause of tangential distortion and its effect on the captured image.

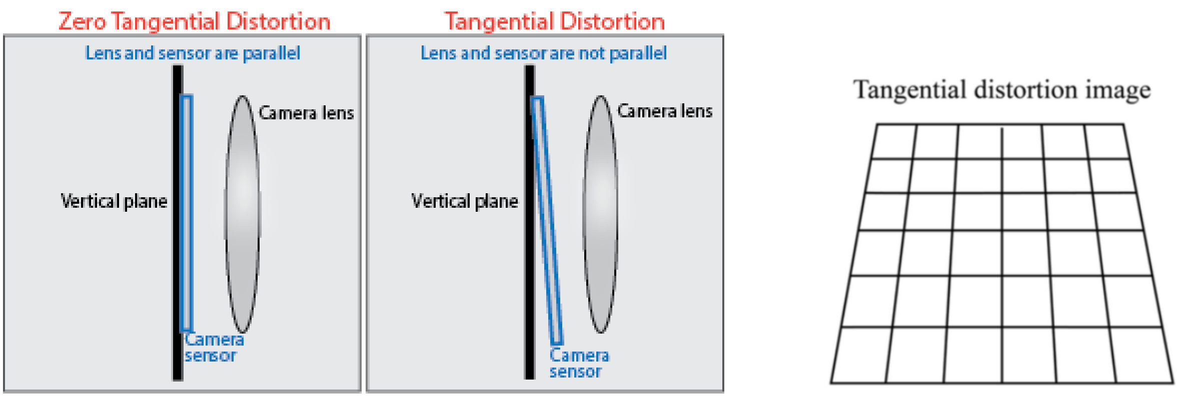

Figure 3: Visualization of tangential distortion. Source: [5], [4].

The formal definition of tangential distortion is given by the following equations [5]:

$$x_{\text{distorted}} = x + \left(2p_1 xy + p_2\left(r^2 + 2x^2\right)\right) \tag{4}$$

$$y_{\text{distorted}} = y + \left(p_1\left(r^2 + 2y^2\right) + 2p_2 xy\right) \tag{5}$$

Here, x and y denote the undistorted normalized image coordinates in the camera coordinate system. The radius r represents the distance of a point from the image center and is defined as:

$$r^2 = x^2 + y^2 \tag{6}$$

The coefficients p_1 and p_2 describe how strong the tangential lens distortion is. This type of distortion is caused by mechanical misalignment of the lens elements relative to the image sensor, resulting in a displacement of image points. Although tangential distortion is typically less pronounced than radial distortion, it can still negatively affect the camera calibration accuracy if not properly modeled.

2.3. Camera Calibration

The goal of camera calibration, also known as camera resectioning, is to estimate the parameters of a camera model. These parameters include the intrinsic parameters, the extrinsic parameters, and the lens distortion coefficients [5]. Knowing these parameters makes it possible to correct distortions in the recorded images and to perform metric measurements in the scene, for example to estimate object sizes or distances from image data.

Camera calibration is required for many tasks in 3D computer vision and photometry. Applications such as depth estimation, 3D reconstruction, or pose estimation rely on a precise mapping from 3D world coordinates to 2D image coordinates [8]. If the calibration is inaccurate, errors can propagate through the processing pipeline and reduce the quality of the final results.

In practice, cameras are often calibrated using targets with known geometric properties. Planar targets, such as checkerboard patterns, are widely used because they are easy to handle and lead to reliable results [9]. Several images of the target are captured from different viewpoints and orientations. Based on these images, the camera parameters can be estimated with relatively small calibration errors. For this reason, target-based calibration methods are commonly used in many computer vision applications.

2.3.1. Calibration Parameters and Error Metrics

- What is camera calibration and which parameters are estimated?
- How are intrinsic and extrinsic parameters defined and interpreted?
- What is the reprojection error and how is it computed?
- Why is the reprojection error commonly used as a calibration quality metric?
- What are the limitations of reprojection-based error measures?

2.3.2. Target-Based Camera Calibration

- What is target-based camera calibration?
- Why are calibration targets (e.g. checkerboards) used?
- How does checkerboard-based calibration work in principle?
- What assumptions are made about the scene and camera setup?
- What are the strengths and limitations of target-based calibration?

2.3.3. Bundle Adjustment

- What is bundle adjustment in the context of camera calibration?
- Which parameters are jointly optimized during bundle adjustment?
- Why does bundle adjustment improve calibration accuracy?
- How is bundle adjustment formulated as an optimization problem?

- What are practical challenges of bundle adjustment in multi-camera systems?

2.4. Light Field Camera Systems

2.4.1. Multi-Camera Light Field Systems

2.4.2. Micro-Lens-Array-Based Light Field Cameras

2.5. State of the Art in Light Field Camera Calibration

3. Material and Method

3.1. Material

3.1.1. Synthetic Data Generation with Blender

3.1.2. Image Sets and Ground Truth Definition

3.1.3. Runtime Environment and Toolchain

3.2. Method

3.2.1. Use Case Description

3.2.2. Requirements Analysis

3.2.3. UML Class Diagram

3.2.4. Health Monitoring and Decision Metrics

3.2.5. State Diagram

4. Results and Evaluation

4.1. Results of Initial Calibration

4.1.1. Intrinsic Accuracy

4.1.2. Extrinsic Accuracy

4.1.3. Reprojection and RMS Error

4.2. Results of LiFCal-Based Recalibration

4.2.1. Reprojection Error after LiFCal

4.2.2. Health Score Evolution

5. Discussion of the Results

5.1. Interpretation of Initial Calibration Results

5.2. Limitations of LiFCal in the Given Setup

5.3. Comparison between Initial Calibration and LiFCal

5.4. Comparison to state of the Art

5.5. Potential Extensions and Improvements

6. Conclusion and Outlook

6.1. Conclusion

6.2. Outlook

A. Appendix

Acknowledgment

References

[1] Elsevier. Radial distortion. https://www.sciencedirect.com/topics/computer-science/radial-distortion, 2022. Definition from ScienceDirect Topics: Computer Science.

[2] Elsevier. Lens distortion. https://www.sciencedirect.com/topics/engineering/lens-distortion, 2024.

[3] Fei-Fei Li et al. Camera models. https://web.stanford.edu/class/cs231a/course_notes/01-camera-models.pdf, 2020.

[4] Shih-Hao Lin, En-Tze Chen, Jia-Jun Xiao, and Ching-Yuan Chang. Stereo digital image correlation (3d-dic) for non-contact measurement and refinement of delta robot arm displacement and jerk. *The International Journal of Advanced Manufacturing Technology*, 133(5-6):2809–2822, July 2024. doi: 10.1007/s00170-024-13957-2. URL https://doi.org/10.1007/s00170-024-13957-2.

[5] MathWorks. Camera calibration. https://de.mathworks.com/help/vision/ug/camera-calibration.html, 2024.

[6] OpenCV Developers. Camera calibration and 3d reconstruction — opencv documentation. OpenCV 4.x Documentation, calib3d module, 2025. URL https://docs.opencv.org/4.x/d9/d0c/group__calib3d.html.

[7] Carlos Ricolfe-Viala and Antonio-José Sánchez-Salmerón. Lens distortion models evaluation. *Applied Optics*, 49(30):5914–5928, 2010. doi: 10.1364/AO.49.005914. URL https://www.researchgate.net/publication/47510646_Lens_distortion_models_evaluation.

[8] Chaehyeon Song, Dongjae Lee, Jongwoo Lim, and Ayoung Kim. Camera calibration via circular patterns: A comprehensive framework with measurement uncertainty and unbiased projection model. *arXiv preprint arXiv:2506.16842*, 2025. URL https://arxiv.org/abs/2506.16842.

[9] Z. Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000. doi: 10.1109/34.888718. URL https://ieeexplore.ieee.org/document/888718.